

Designing the Entity Relationship (ER) Diagram and Relational Schema

Chapter 3

3.1. Introduction

The Entity Relationship (ER) Diagram is a graphical representation of the database that illustrates the entities involved in the system and the relationships between them. It serves as the blueprint for designing the database before implementation. Creating an ER diagram helps identify the required tables, attributes, primary keys, foreign keys, and relationships, making the database more organized, scalable, and easier to maintain.

For the IKEA E-Commerce Database Management System, the ER diagram was designed to represent the complete workflow of an online furniture shopping platform, including customer management, product catalog, shopping cart, orders, payments, inventory, shipment, reviews, and AI-based recommendations.

3.2. Objectives

The objectives of designing the ER Diagram are:

- To identify all the entities required in the system.
- To establish relationships among entities.
- To minimize data redundancy using database normalization.
- To define primary and foreign keys.
- To provide a clear structure before database implementation.
- To create a scalable and maintainable database architecture.

3.3. Entity Identification

The following entities were identified for the IKEA E-Commerce Database:

- Customer
- Customer_Address
- Product
- Category

- Supplier
- Warehouse
- Inventory
- Cart
- Cart_Item
- Wishlist
- Wishlist_Item
- Order
- Order_Item
- Coupon
- Payment
- Shipment
- Return_Request
- Review
- Product_Image
- Notification
- Order_Status_History
- AI_Recommendation
- AI_Search_History
- Analytics
- Admin

3.4. Relationship Design

The relationships between entities were identified based on real-world business requirements.

Some important relationships include:

- One customer can have multiple addresses.
- One customer can place multiple orders.
- One category contains multiple products.
- One supplier supplies multiple products.

- One warehouse stores multiple products through the Inventory table.
- One cart contains multiple cart items.
- One wishlist contains multiple wishlist items.
- One order contains multiple order items.
- One product can receive multiple reviews.
- One customer can receive multiple AI recommendations.

These relationships ensure proper organization of data and reduce redundancy.

3.5. Relational Schema

The ER Diagram was converted into a relational schema by representing each entity as a table. Every table contains:

- Primary Key
- Foreign Keys (where applicable)
- Attributes

The relational schema provides the logical structure of the database before writing SQL queries.

3.6. Database Improvements

To make the project closer to a real-world e-commerce platform, several advanced modules were added.

Customer Module

- Customer
- Customer_Address
- Wishlist
- Wishlist_Item
- Cart
- Cart_Item

Product Module

- Product
- Category

- Supplier
- Warehouse
- Inventory
- Product_Image

Order Module

- Order
- Order_Item
- Coupon
- Payment
- Shipment
- Return_Request
- Order_Status_History

AI Module

- AI_Recommendation
- AI_Search_History
- Analytics

Administration Module

- Admin
- Notification
- Review

These additions improve scalability and better represent enterprise-level database design.

3.7. Tools Used

- **DBDiagram.io** – Designing the ER Diagram.
- **GitHub** – Version control and project documentation.
- **Microsoft Word** – Preparing project documentation.
- **Oracle SQL** (*next phase*) – Database implementation.

3.8. Learning Outcome

Through this phase of the project, the following concepts were understood:

- Entity Identification
- Attributes
- Primary Keys
- Foreign Keys
- One-to-One Relationships
- One-to-Many Relationships
- Many-to-Many Relationships
- Database Normalization
- ER Diagram Design
- Relational Schema Design
- Real-world E-Commerce Database Architecture

3.9. Conclusion

The ER Diagram and relational schema provide a strong foundation for the IKEA E-Commerce Database Management System. The database has been designed following database normalization principles and industry practices, making it scalable, organized, and suitable for future implementation using SQL. This design supports essential e-commerce operations such as customer management, product catalog, inventory tracking, order processing, payments, shipment, reviews, and AI-driven recommendations.